

RoboSoccer

Bluetooth-Controlled Soccer Robots

Project Proposal

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Why This Project

With the FIFA World Cup 2026 around the corner, football fever is everywhere – and we wanted to bring that same excitement into our lab. Instead of another routine sensor-and-motor exercise, we chose to build something we would genuinely enjoy designing, testing, and playing with.

- **Entertainment & refreshment:** a physical, competitive game that turns lab time into something fun rather than another checklist task.
- **Inspired by FIFA WC 2026:** the global excitement around football this year directly shaped our choice of a soccer-robot theme.
- **Hands-on learning:** still exercises core embedded concepts (UART, PWM, timers, interrupts) – just wrapped in a project we're actually excited to finish.

Objectives

- Design two independent 4-wheel drive robot cars, each controlled by a mobile app over Bluetooth.
- Implement smooth motor control using PWM through TB6612FNG drivers.
- Add a kicker mechanism using a servo motor.
- Implement a laser-based goal detection system at each goalpost.
- Demonstrate a complete two-player robo-soccer match as final deliverable.

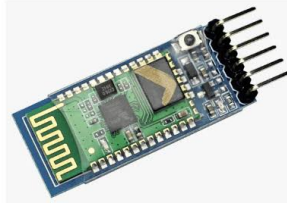
Equipment – ATmega32

Main microcontroller, motor & servo control, sensor input.



Equipment – HC-05 Bluetooth Module

Receives drive/kick commands from the Android app.



Equipment – TB6612FNG Motor Driver

Drives the 4 DC motors (left pair / right pair).



Equipment – 6V Geared DC Motors, TT-type

Drive the robot's wheels.



Equipment – Servo Motor, SG90/MG90S/MG996R

Rotates the kicking arm to shoot the ball.



Equipment – Robot Chassis + Wheels

Base frame and locomotion.



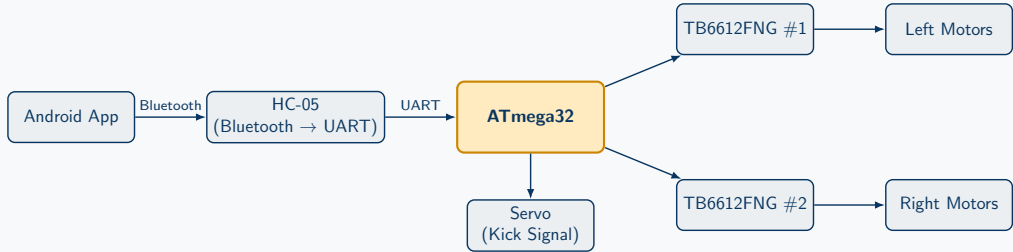
Equipment – Additional Components

- **Servo Horn & Screws, Kicking Arm** – converts servo rotation into a kicking strike.
- **U-shaped Ball Holder** – holds the ball in front of the robot before the kick.
- **Breadboard & Jumper Wires**
- **Laser Module + Laser Receiver** – detects ball crossing the goal line.

Equipment – Software / Tools

- **Atmel Studio / AVR-GCC** – firmware development for ATmega32.
- **AVRDUDE** – flashing the microcontroller.
- **Proteus** – circuit simulation and PCB layout.
- **Mobile App** – For Bluetooth control.

System Design – Control Connections



System Design – Component Roles

- **Android App**: sends drive/kick commands over Bluetooth serial.
- **HC-05**: wireless UART bridge to ATmega32.
- **ATmega32**: decodes commands, generates PWM, manages sensor interrupts.
- **TB6612FNG**: converts PWM + direction signals into motor drive current.
- **DC Motors**: provide locomotion for the 4-wheel chassis.
- **Servo + Kicking Arm**: strikes the ball on command.
- **Laser + Receiver**: detects the ball breaking the beam at the goal mouth.
- **2S Battery + BMS**: raw power source with charge/discharge protection.
- **LM2596**: regulated 5V supply for ATmega32, HC-05, and (5V) servo.
- **Main Switch**: master on/off for the whole system.

Key ATmega32 Pins / Interfaces

- **UART (RXD/TXD):** serial link to HC-05 for app commands.
- **Timer0/Timer1 PWM (OC0, OC1A/OC1B):** speed control for drive motors and servo signal.
- **GPIO Port A/B:** direction control pins (AIN1/AIN2, BIN1/BIN2) for TB6612FNG.
- **External Interrupt (INT0/INT1):** goal-detection trigger from laser receiver.
- **ADC channel:** battery voltage monitoring (optional).

Software Features – Command Protocol

App Layout

The Android app has six buttons; each press sends a single ASCII character over Bluetooth, which the ATmega32 receives via UART and maps to an action.

Button	Char	ATmega32 Action
Forward	F	Both drivers forward, equal PWM duty.
Backward	B	Both drivers reverse, equal PWM duty.
Left	L	Left motors reverse / right motors forward (or reduced left speed).
Right	R	Right motors reverse / left motors forward (or reduced right speed).
Stop	S	All motor PWM set to 0.
Kick	K	Servo sweeps to kick angle, then returns to rest.

Role of the ATmega32

Core Responsibilities

The ATmega32 is the central controller for each car – it is not a passive relay but performs all real-time decision making and actuation.

- Parses Bluetooth commands received via UART from the HC-05 module.
- Generates PWM signals (Timer0/Timer1) to control 4 drive motors through two TB6612FNG drivers.
- Generates a separate PWM signal to position the servo-driven kicker.
- Uses external interrupts to instantly detect a goal event from the laser receiver.
- Uses GPIO pins to set motor direction (forward/reverse/turn logic).
- Optionally uses ADC to monitor battery voltage and trigger a low-battery alert.

Implementation Plan – Expected Outcome

Final Outcome

Two independently Bluetooth-controlled 4-wheel robot cars capable of driving, kicking a ball, and automatically detecting goals via a laser-beam sensor – forming a complete two-player robo-soccer match.

Implementation Plan – 5 Week Timeline

Week	Milestone
Week 1	Component procurement, chassis assembly, basic ATmega32 + motor driver testing.
Week 2	HC-05 Bluetooth interfacing, UART command protocol design, Mobile App configuration
Week 3	Project Update 1: single car drives via app control (forward/back/turn) with PWM speed control.
Week 4	Servo kicker integration, laser goal-detection module testing, dual-car synchronization.
Week 5	Final Submission: full two-car match demo, goal detection, bug fixing, documentation.

Milestones & Dependencies

- Motor driver testing depends on chassis assembly being complete.
- Bluetooth command protocol must be finalized before app development.
- Goal detection integration depends on laser module calibration.
- Final demo depends on both cars being functionally independent and tested together.

Use of Other Controllers

Note

This project does not require an additional microcontroller (e.g. ESP32, Arduino, Raspberry Pi). The ATmega32 alone handles motor control, servo actuation, Bluetooth communication, and goal-detection.

Testing Strategy

- Unit test each module individually: motor driver response, servo angle accuracy, Bluetooth range/latency.
- Integration testing: full command-to-motion latency measurement.
- Field testing: goal-detection reliability under different lighting/ball speeds.
- Stress testing: continuous operation time on battery power.

Success Criteria

Essential (Must-Have)

- Reliable Bluetooth control of both cars.
- Smooth forward/reverse/turn motion via PWM.
- Functional servo kicker.
- Accurate goal detection via laser sensor.

Optional (Nice-to-Have)

- Battery-level monitoring and alert.
- Score display/counter.
- Adjustable motor speed profiles.
- LED indicators for game status.

Challenges

- Ensuring low-latency, interference-free dual Bluetooth links during simultaneous play.
- Precise motor speed matching for straight-line driving (motor tolerance variation).
- Reliable laser-receiver triggering under ambient light interference.
- Power budget management for motors, servo, and logic on a single 2S battery pack.

Future Scope

- Implement autonomous ball detection and tracking using a camera or vision sensor.
- Add obstacle detection and avoidance using ultrasonic or infrared sensors.
- Integrate wheel encoders for accurate speed and distance control.
- Upgrade to a custom PCB to improve reliability and reduce wiring complexity.
- Improve the kicking mechanism using a solenoid or spring-loaded actuator for stronger shots.

Thank You